Econometrics: Methods and Applications

Test Exercise 2: Small-and-Large Models (Matrix Verification)

(a) Prove that $E(b_R) = \beta_1 + P\beta_2$

Proof:

Let the true population regression model (unrestricted large model) be:

$y = X_1\beta_1 + X_2\beta_2 + \varepsilon$, where $E(\varepsilon) = 0$.

The OLS estimator for the restricted small model (omitting X_2) is:

$b_R = (X_1'X_1)^{-1}X_1'y$

Substituting the true value of y into the formula:

$b_R = (X_1'X_1)^{-1}X_1'(X_1\beta_1 + X_2\beta_2 + \varepsilon)$

$b_R = (X_1'X_1)^{-1}X_1'X_1\beta_1 + (X_1'X_1)^{-1}X_1'X_2\beta_2 + (X_1'X_1)^{-1}X_1'\varepsilon$

Since $(X_1'X_1)^{-1}X_1'X_1 = I$ and by definition $P = (X_1'X_1)^{-1}X_1'X_2$:

$b_R = \beta_1 + P\beta_2 + (X_1'X_1)^{-1}X_1'\varepsilon$

Taking mathematical expectations on both sides, assuming X is non-stochastic and $E(\varepsilon) = 0$:

$E(b_R) = E[\beta_1 + P\beta_2 + (X_1'X_1)^{-1}X_1'\varepsilon]$

$E(b_R) = \beta_1 + P\beta_2 + (X_1'X_1)^{-1}X_1'E(\varepsilon)$

$E(b_R) = \beta_1 + P\beta_2$ [Proved]

(b) Prove that $var(b_R) = \sigma^2(X_1'X_1)^{-1}$

Proof:

From part (a), we know that:

$b_R - E(b_R) = (X_1'X_1)^{-1}X_1'\varepsilon$

Using the definition of the variance-covariance matrix for a random vector matrix product:

$var(b_R) = E[(b_R - E(b_R))(b_R - E(b_R))']$

$var(b_R) = E[((X_1'X_1)^{-1}X_1'\varepsilon)((X_1'X_1)^{-1}X_1'\varepsilon)']$

$var(b_R) = E[(X_1'X_1)^{-1}X_1'\varepsilon\varepsilon'X_1(X_1'X_1)^{-1}]$

Factoring out the deterministic matrix components and substituting $E(\varepsilon\varepsilon') = \sigma^2 I$ (under homoscedasticity and no autocorrelation):

$var(b_R) = (X_1'X_1)^{-1}X_1'\,[\sigma^2 I]\,X_1(X_1'X_1)^{-1}$

$var(b_R) = \sigma^2 (X_1'X_1)^{-1}\,(X_1'X_1)\,(X_1'X_1)^{-1}$

$var(b_R) = \sigma^2(X_1'X_1)^{-1}$ [Proved]

(c) Prove that $b_R = b_1 + Pb_2$

Proof:

From the unrestricted large model OLS estimation, the actual sample vector y can be split into its fitted values and OLS residuals vector e:

$y = X_1 b_1 + X_2 b_2 + e$

By the fundamental first-order conditions of OLS, the residuals are perfectly orthogonal to the included regressors, implying:

$X_1'e = 0$

Now, substitute this sample decomposition of y into the restricted OLS estimator expression:

$b_R = (X_1'X_1)^{-1}X_1'y$

$b_R = (X_1'X_1)^{-1}X_1'(X_1 b_1 + X_2 b_2 + e)$

$b_R = (X_1'X_1)^{-1}X_1'X_1 b_1 + (X_1'X_1)^{-1}X_1'X_2 b_2 + (X_1'X_1)^{-1}X_1'e$

Simplifying using $(X_1'X_1)^{-1}X_1'X_1 = I$, $P = (X_1'X_1)^{-1}X_1'X_2$, and $X_1'e = 0$:

$b_R = I\, b_1 + P\, b_2 + 0$

$b_R = b_1 + Pb_2$ [Proved]

(d) Argue that the columns of P are obtained by regressing Age, Educ, and Parttime on a Constant and Female

Argument:

The matrix P is defined as $P = (X_1'X_1)^{-1}X_1'X_2$.

Here, X_1 contains two columns: a constant column (vector of 1s) and the 'Female' indicator variable. X_2 contains three columns representing the variables 'Age', 'Educ', and 'Parttime'.

We can partition X_2 by columns: $X_2 = [\, x_{Age} \mid x_{Educ} \mid x_{Parttime} \,]$. Then:

$P = [\, (X_1'X_1)^{-1}X_1'x_{Age} \mid (X_1'X_1)^{-1}X_1'x_{Educ} \mid (X_1'X_1)^{-1}X_1'x_{Parttime} \,]$

Each column vector template $(X_1'X_1)^{-1}X_1'x_j$ represents exactly the standard OLS formula for the vector of coefficients obtained by regressing the variable x_j onto the matrix of regressors X_1 (Constant and Female). Therefore, the columns of P are precisely the intercept and slope coefficients from these three individual auxiliary regressions.

(e) Determine the values of P from the results in Lecture 2.1

Using the dataset and results discussed in Lecture 2.1, running the three auxiliary regressions yields the following intercept and slope values for matrix P:

Regressors	Age	Educ	Parttime
Constant (Intercept)	40.05	2.26	0.20
Female (Slope)	-0.11	-0.49	0.25

(f) Check the numerical validity of the result in part (c)

From the full unrestricted model regression (Lecture 2.5), the estimated coefficients are:
b_1 (Intercept) = 3.0527, b_1 (Female) = -0.0411
b_2 (Age) = 0.0306, b_2 (Educ) = 0.2332, b_2 (Parttime) = -0.3654

We compute the right-hand side $b_1 + Pb_2$ for both parameters:

1. Intercept component:
3.0527 + (40.05 × 0.0306) + (2.26 × 0.2332) + (0.20 × -0.3654)
= 3.0527 + 1.2255 + 0.5270 - 0.0731 = **4.7321**

2. Female coefficient component:
-0.0411 + (-0.11 × 0.0306) + (-0.49 × 0.2332) + (0.25 × -0.3654)
= -0.0411 - 0.0034 - 0.1143 - 0.0914 = **-0.2502**

From the short restricted model in Lecture 2.1, the OLS estimates are exactly:
b_R (Intercept) = **4.73** and b_R (Female) = **-0.25**.

The values match up perfectly to two decimal places (allowing for minor rounding differences in the lecture slides), verifying the numerical validity of the identity $\mathbf{b_R = b_1 + Pb_2}$.